

Session

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Math Coaching Assistant



Adult tickets to a show cost \$12 each, and student tickets cost \$8 each. A total of 25 tickets were sold, generating \$248 in revenue. How many adult tickets and student tickets were sold?



To solve this, we first need to translate the problem into equations. Let a represent the number of adult tickets and s represent the number of student tickets. Based on the information that 25 tickets were sold in total, which equation is correct?



$a + s = 25$



That's correct. Now we need a second equation to represent the total revenue of \$248. Given that adult tickets cost \$12 and student tickets cost \$8, which equation represents the total money collected?



$a + s = 248$



That is incorrect because $a + s$ represents the total number of tickets, which we already know is 25. To find the revenue, we must multiply the number of tickets by their respective prices. If one adult ticket costs \$12, how much money is made from selling a adult tickets?



12a

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12a



Exactly, 12a represents the revenue from adult tickets. Now, considering that student tickets cost \$8 each and the total revenue is \$248, what is the complete second equation?

 $12a + 8s = 248$ 

Correct. Our system is $a + s = 25$ and $12a + 8s = 248$. If we use substitution and solve the first equation for s to get $s = 25 - a$, which expression represents the revenue equation after substituting?

 $12a + 8(25 - a) = 248$ 

That's correct. Now, let's simplify the equation by distributing the 8 to the terms inside the parentheses. What does the equation look like after this step?

 $12a + 200 - 8a = 248$ 

That is correct. Now, let's combine the like terms (the terms with 'a') on the left side of the equation $12a + 200 - 8a = 248$. What is the resulting equation?

 $4a + 200 = 248$

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$$4a + 200 = 248$$



That's correct. Now, to isolate the term with 'a', we need to subtract 200 from both sides of the equation $4a + 200 = 248$. What is the resulting equation?



$$4a = 48$$



That's correct. Now, divide both sides by 4 to find the value of a. What is the value of a in the equation $4a = 48$?



$$a = 12$$



That's correct, so there were 12 adult tickets sold. Now we need to find the number of student tickets using our first equation, $12 + s = 25$. What is the value of s?



$$s = 13$$



That is correct. You have successfully solved the problem by first translating the word problem into a system of equations: one for the total number of tickets ($a + s = 25$) and one for the total revenue ($12a + 8s = 248$). You then used substitution to eliminate one variable, simplified the resulting equation to find that 12 adult tickets were sold, and finally used that value to determine that 13 student tickets were sold.

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